

Saber Ayatollahi, CFA, MSc

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SUMMARY

Quantitative risk and model-development professional with ~7 years across market and balance-sheet risk, derivatives valuation, and model governance, targeting an Associate Director, Global Risk Analytics role. Holds formal sign-off authority on valuation, sensitivity, and ALM outputs for multi-asset institutional portfolios (\$5–25bn per engagement; \$50bn+ cumulative) at Moody's Analytics, developing, benchmarking, and validating pricing and scenario models across rates, FX, and inflation. Strong in Python, MATLAB, SQL, stochastic / Monte Carlo simulation, and statistics, with applied knowledge of market risk, counterparty (CCR), and CCAR-style stress testing. CFA charterholder with dual MSc.

CORE SKILLS

Market & Counterparty Credit Risk (CCR) · Derivatives Valuation – Rates, FX, Inflation · Stress Testing & Scenario Analysis (CCAR-style) · Quantitative Model Development & Methodology · Stochastic & Monte Carlo Simulation · Statistics · Model Governance & Independent Validation · Python · MATLAB · SQL · Excel · Yield Curve Construction & Calibration · Risk Reporting & Regulator / Executive Communication

EXPERIENCE

Moody's Analytics – Asset Management & Buy-Side Solutions

Asst. Director – Modelling Services | Toronto, May 2023 – Present

Model Development, Valuation & Risk Oversight

- Hold formal sign-off authority for valuation, sensitivity, and ALM outputs across multi-asset institutional portfolios (\$5–25bn per engagement; \$50bn+ cumulative), independently reviewing curve construction, spread calibration, and cross-asset interactions before release.
- Develop and validate derivatives pricing and sensitivity models across rates, FX, and inflation, ensuring methodologies are mathematically and economically defensible and consistent at portfolio-level aggregates.
- Build independently developed benchmark models to challenge production outputs and provide implementation detail and acceptance criteria to technology teams for deployment.
- Oversee interest-rate and market exposure under parallel and non-parallel shock scenarios, aligned with IRRBB and Basel Committee market-risk frameworks.
- Document model methodologies, assumptions, limitations, and testing results to support internal validation review and approval.

Scenario, Stress & Liquidity Analytics

- Led design of an enterprise multi-asset cash flow projection and scenario engine (base, stress, reverse-stress) — applicable to CCAR-style forward-looking stress assessment.
- Architected configurable time-bucketed liquidity gap analytics (T+1 through multi-year) supporting funding, refinancing, and capital-adequacy assessment.
- Developed tools to monitor model performance, assumptions, and limitations on an ongoing basis, investigating and remediating issues identified through monitoring or independent validation.
- Re-engineered manual spreadsheet workflows into auditable Python pipelines; built agentic AI workflows (Claude Code, Cursor IDE) cutting development and validation cycle time ~30–40%.

Governance & Stakeholder Communication

- Operate within a formal model-governance framework, reviewing assumptions and stress methodologies and escalating outputs lacking mathematical or economic defensibility.
- Explain complex modelling concepts and scenario impacts to senior management and clients' Heads of Risk, translating quantitative outputs into decision-ready narratives.
- Collaborate with Risk, Product, Technology, and Data teams to deliver production-ready, controlled, and auditable analytics.

Client Service Specialist – ALM & Risk Analytics | Toronto, May 2022 – May 2023

- Led onboarding and implementation for U.S. and Canadian pension funds, asset managers, and consulting firms migrating to the PFaroe analytics platform.
- Served as technical liaison between client investment teams and product owners, translating requirements into development requests and validating model outputs post-deployment.
- Contributed to the successful migration of all assigned client accounts from the legacy Calypso platform to the PFaroe PM platform.

Ernst & Young (EY)***Senior Consultant – IFRS & Financial Risk Transformation | Toronto, Sep 2021 – Apr 2022***

- Delivered IFRS 17 and IFRS 9 transformation projects for Canadian and international insurance clients, aligning regulatory requirements with model and system implementation.
- Advised actuarial, finance, IT, and risk functions; contributed to governance documentation and regulatory-readiness processes for financial institutions.

Ortec Finance Canada***Senior Consultant – Asset-Liability Management & Risk | Toronto, Feb 2019 – Sep 2021***

- Delivered ALM studies for pension funds and institutional investors, analyzing funding ratios, duration mismatch, and long-term balance-sheet risk exposures.
- Built and interpreted stochastic economic scenario generators (Monte Carlo) to assess interest-rate risk and funding volatility under base-case and stressed environments.
- Conducted liability-driven investment (LDI) analysis, evaluating trade-offs between return objectives, duration alignment, and liquidity considerations.
- Presented findings to investment committees and senior stakeholders to support strategic allocation decisions.

EDUCATION & CREDENTIALS

- Chartered Financial Analyst (CFA) – CFA Institute, 2024
- MSc, Financial Modelling – Western University, 2018
- MSc, Chemical Engineering (Full Scholarship) – Western University, 2016